

NAME

xz, **unxz**, **xzcat**, **lzma**, **unlzma**, **lzcat** – Compress or decompress **.xz** and **.lzma** files

SYNOPSIS

xz [*option...*] [*file...*]

COMMAND ALIASES

unxz is equivalent to **xz --decompress**.

xzcat is equivalent to **xz --decompress --stdout**.

lzma is equivalent to **xz --format=lzma**.

unlzma is equivalent to **xz --format=lzma --decompress**.

lzcat is equivalent to **xz --format=lzma --decompress --stdout**.

When writing scripts that need to decompress files, it is recommended to always use the name **xz** with appropriate arguments (**xz -d** or **xz -dc**) instead of the names **unxz** and **xzcat**.

DESCRIPTION

xz is a general-purpose data compression tool with command line syntax similar to **gzip**(1) and **bzip2**(1). The native file format is the **.xz** format, but the legacy **.lzma** format used by LZMA Utils and raw compressed streams with no container format headers are also supported. In addition, decompression of the **.lz** format used by **lzip** is supported.

xz compresses or decompresses each *file* according to the selected operation mode. If no *files* are given or *file* is **-**, **xz** reads from standard input and writes the processed data to standard output. **xz** will refuse (display an error and skip the *file*) to write compressed data to standard output if it is a terminal. Similarly, **xz** will refuse to read compressed data from standard input if it is a terminal.

Unless **--stdout** is specified, *files* other than **-** are written to a new file whose name is derived from the source *file* name:

- When compressing, the suffix of the target file format (**.xz** or **.lzma**) is appended to the source filename to get the target filename.
- When decompressing, the **.xz**, **.lzma**, or **.lz** suffix is removed from the filename to get the target filename. **xz** also recognizes the suffixes **.txz** and **.tlz**, and replaces them with the **.tar** suffix.

If the target file already exists, an error is displayed and the *file* is skipped.

Unless writing to standard output, **xz** will display a warning and skip the *file* if any of the following applies:

- *File* is not a regular file. Symbolic links are not followed, and thus they are not considered to be regular files.
- *File* has more than one hard link.
- *File* has **setuid**, **setgid**, or sticky bit set.
- The operation mode is set to compress and the *file* already has a suffix of the target file format (**.xz** or **.txz** when compressing to the **.xz** format, and **.lzma** or **.tlz** when compressing to the **.lzma** format).

- The operation mode is set to decompress and the *file* doesn't have a suffix of any of the supported file formats (**.xz**, **.txz**, **.lzma**, **.tlz**, or **.lz**).

After successfully compressing or decompressing the *file*, **xz** copies the owner, group, permissions, access time, and modification time from the source *file* to the target file. If copying the group fails, the permissions are modified so that the target file doesn't become accessible to users who didn't have permission to access the source *file*. **xz** doesn't support copying other metadata like access control lists or extended attributes yet.

Once the target file has been successfully closed, the source *file* is removed unless **--keep** was specified. The source *file* is never removed if the output is written to standard output or if an error occurs.

Sending **SIGINFO** or **SIGUSR1** to the **xz** process makes it print progress information to standard error. This has only limited use since when standard error is a terminal, using **--verbose** will display an automatically updating progress indicator.

Memory usage

The memory usage of **xz** varies from a few hundred kilobytes to several gigabytes depending on the compression settings. The settings used when compressing a file determine the memory requirements of the decompressor. Typically the decompressor needs 5 % to 20 % of the amount of memory that the compressor needed when creating the file. For example, decompressing a file created with **xz -9** currently requires 65 MiB of memory. Still, it is possible to have **.xz** files that require several gigabytes of memory to decompress.

Especially users of older systems may find the possibility of very large memory usage annoying. To prevent uncomfortable surprises, **xz** has a built-in memory usage limiter, which is disabled by default. While some operating systems provide ways to limit the memory usage of processes, relying on it wasn't deemed to be flexible enough (for example, using **ulimit(1)** to limit virtual memory tends to cripple **mmap(2)**).

The memory usage limiter can be enabled with the command line option **--memlimit=limit**. Often it is more convenient to enable the limiter by default by setting the environment variable **XZ_DEFAULTS**, for example, **XZ_DEFAULTS=--memlimit=150MiB**. It is possible to set the limits separately for compression and decompression by using **--memlimit-compress=limit** and **--memlimit-decompress=limit**. Using these two options outside **XZ_DEFAULTS** is rarely useful because a single run of **xz** cannot do both compression and decompression and **--memlimit=limit** (or **-M limit**) is shorter to type on the command line.

If the specified memory usage limit is exceeded when decompressing, **xz** will display an error and decompressing the file will fail. If the limit is exceeded when compressing, **xz** will try to scale the settings down so that the limit is no longer exceeded (except when using **--format=raw** or **--no-adjust**). This way the operation won't fail unless the limit is very small. The scaling of the settings is done in steps that don't match the compression level presets, for example, if the limit is only slightly less than the amount required for **xz -9**, the settings will be scaled down only a little, not all the way down to **xz -8**.

Concatenation and padding with .xz files

It is possible to concatenate **.xz** files as is. **xz** will decompress such files as if they were a single **.xz** file.

It is possible to insert padding between the concatenated parts or after the last part. The padding must consist of null bytes and the size of the padding must be a multiple of four bytes. This can be useful, for example, if the **.xz** file is stored on a medium that measures file

sizes in 512-byte blocks.

Concatenation and padding are not allowed with **.lzma** files or raw streams.

OPTIONS

Integer suffixes and special values

In most places where an integer argument is expected, an optional suffix is supported to easily indicate large integers. There must be no space between the integer and the suffix.

KiB Multiply the integer by 1,024 (2^{10}). **Ki**, **k**, **kB**, **K**, and **KB** are accepted as synonyms for **KiB**.

MiB Multiply the integer by 1,048,576 (2^{20}). **Mi**, **m**, **M**, and **MB** are accepted as synonyms for **MiB**.

GiB Multiply the integer by 1,073,741,824 (2^{30}). **Gi**, **g**, **G**, and **GB** are accepted as synonyms for **GiB**.

The special value **max** can be used to indicate the maximum integer value supported by the option.

Operation mode

If multiple operation mode options are given, the last one takes effect.

-z, --compress

Compress. This is the default operation mode when no operation mode option is specified and no other operation mode is implied from the command name (for example, **unxz** implies **--decompress**).

After successful compression, the source file is removed unless writing to standard output or **--keep** was specified.

-d, --decompress, --uncompress

Decompress. After successful decompression, the source file is removed unless writing to standard output or **--keep** was specified.

-t, --test

Test the integrity of compressed *files*. This option is equivalent to **--decompress --stdout** except that the decompressed data is discarded instead of being written to standard output. No files are created or removed.

-l, --list

Print information about compressed *files*. No uncompressed output is produced, and no files are created or removed. In list mode, the program cannot read the compressed data from standard input or from other unseekable sources.

The default listing shows basic information about *files*, one file per line. To get more detailed information, use also the **--verbose** option. For even more information, use **--verbose** twice, but note that this may be slow, because getting all the extra information requires many seeks. The width of verbose output exceeds 80 characters, so piping the output to, for example, **less -S** may be convenient if the terminal isn't wide enough.

The exact output may vary between **xz** versions and different locales. For machine-readable output, **--robot --list** should be used.

Operation modifiers

-k, --keep

Don't delete the input files.

Since **xz** 5.2.6, this option also makes **xz** compress or decompress even if the input is a symbolic link to a regular file, has more than one hard link, or has the **setuid**, **setgid**, or sticky bit set. The **setuid**, **setgid**, and sticky bits are not copied to the target file. In earlier versions this was only done with **--force**.

-f, --force

This option has several effects:

- If the target file already exists, delete it before compressing or decompressing.
- Compress or decompress even if the input is a symbolic link to a regular file, has more than one hard link, or has the **setuid**, **setgid**, or sticky bit set. The **setuid**, **setgid**, and sticky bits are not copied to the target file.
- When used with **--decompress --stdout** and **xz** cannot recognize the type of the source file, copy the source file as is to standard output. This allows **xzcat --force** to be used like **cat(1)** for files that have not been compressed with **xz**. Note that in future, **xz** might support new compressed file formats, which may make **xz** decompress more types of files instead of copying them as is to standard output. **--format=format** can be used to restrict **xz** to decompress only a single file format.

-c, --stdout, --to-stdout

Write the compressed or decompressed data to standard output instead of a file. This implies **--keep**.

--single-stream

Decompress only the first **.xz** stream, and silently ignore possible remaining input data following the stream. Normally such trailing garbage makes **xz** display an error.

xz never decompresses more than one stream from **.lzma** files or raw streams, but this option still makes **xz** ignore the possible trailing data after the **.lzma** file or raw stream.

This option has no effect if the operation mode is not **--decompress** or **--test**.

Since **xz** 5.7.1alpha, **--single-stream** implies **--keep**.

--no-sparse

Disable creation of sparse files. By default, if decompressing into a regular file, **xz** tries to make the file sparse if the decompressed data contains long sequences of binary zeros. It also works when writing to standard output as long as standard output is connected to a regular file and certain additional conditions are met to make it safe. Creating sparse files may save disk space and speed up the decompression by reducing the amount of disk I/O.

-S *.suf*, **--suffix=***.suf*

When compressing, use *.suf* as the suffix for the target file instead of **.xz** or **.lzma**. If not writing to standard output and the source file already has the suffix *.suf*, a warning is displayed and the file is skipped.

When decompressing, recognize files with the suffix *.suf* in addition to files with the **.xz**, **.txz**, **.lzma**, **.tlz**, or **.lz** suffix. If the source file has the suffix *.suf*, the suffix is removed to get the target filename.

When compressing or decompressing raw streams (**--format=raw**), the suffix must always be specified unless writing to standard output, because there is no default suffix for raw streams.

--files[=file]

Read the filenames to process from *file*; if *file* is omitted, filenames are read from standard input. Filenames must be terminated with the newline character. A dash (-) is taken as a regular filename; it doesn't mean standard input. If filenames are given also as command line arguments, they are processed before the filenames read from *file*.

--files0[=file]

This is identical to **--files[=file]** except that each filename must be terminated with the null character.

Basic file format and compression options

-F *format*, **--format=***format*

Specify the file *format* to compress or decompress:

auto This is the default. When compressing, **auto** is equivalent to **xz**. When decompressing, the format of the input file is automatically detected. Note that raw streams (created with **--format=raw**) cannot be auto-detected.

xz Compress to the **.xz** file format, or accept only **.xz** files when decompressing.

lzma, alone

Compress to the legacy **.lzma** file format, or accept only **.lzma** files when decompressing. The alternative name **alone** is provided for backwards compatibility with LZMA Utils.

lzip Accept only **.lz** files when decompressing. Compression is not supported.

The **.lz** format versions 0 and 1 are supported. Version 0 files were produced by **lzip** 1.3 and older. Such files aren't common but may be found from file archives as a few source packages were released in this format. People might have old personal files in this format too. Decompression support for the format version 0 was removed in **lzip** 1.18. **lzip** 1.4 and later create files in the format version 1.

raw Compress or uncompress a raw stream (no headers). This is meant for advanced users only. To decode raw streams, you need use **--format=raw** and explicitly specify the filter chain, which normally would have been stored in the container headers.

-C check, --check=check

Specify the type of the integrity check. The check is calculated from the uncompressed data and stored in the **.xz** file. This option has an effect only when compressing into the **.xz** format; the **.lzma** format doesn't support integrity checks. The integrity check (if any) is verified when the **.xz** file is decompressed.

Supported *check* types:

none Don't calculate an integrity check at all. This is usually a bad idea. This can be useful when integrity of the data is verified by other means anyway.

crc32 Calculate CRC32 using the polynomial from IEEE-802.3 (Ethernet).

crc64 Calculate CRC64 using the polynomial from ECMA-182. This is the default, since it is slightly better than CRC32 at detecting damaged files and the speed difference is negligible.

sha256

Calculate SHA-256. This is somewhat slower than CRC32 and CRC64.

Integrity of the **.xz** headers is always verified with CRC32. It is not possible to change or disable it.

--ignore-check

Don't verify the integrity check of the compressed data when decompressing. The CRC32 values in the **.xz** headers will still be verified normally.

Do not use this option unless you know what you are doing. Possible reasons to use this option:

- Trying to recover data from a corrupt **.xz** file.
- Speeding up decompression. This matters mostly with SHA-256 or with files that have compressed extremely well. It's recommended to not use this option for this purpose unless the file integrity is verified externally in some other way.

-0 ... -9

Select a compression preset level. The default is **-6**. If multiple preset levels are specified, the last one takes effect. If a custom filter chain was already specified, setting a compression preset level clears the custom filter chain.

The differences between the presets are more significant than with **gzip(1)** and **bzip2(1)**. The selected compression settings determine the memory requirements of the decompressor, thus using a too high preset level might make it painful to decompress the file on an old system with little RAM. Specifically, **it's not a good idea to blindly use -9 for everything** like it often is with **gzip(1)** and **bzip2(1)**.

-0 ... -3

These are somewhat fast presets. **-0** is sometimes faster than **gzip -9** while compressing much better. The higher ones often have speed comparable to **bzip2(1)** with comparable or better compression ratio, although the results depend a lot on the type of data being compressed.

-4 ... -6

Good to very good compression while keeping decompressor memory usage reasonable even for old systems. **-6** is the default, which is usually a good choice for distributing files that need to be decompressible even on systems with only 16 MiB RAM. (**-5e** or **-6e** may be worth considering too. See **--extreme**.)

-7 ... -9

These are like **-6** but with higher compressor and decompressor memory requirements. These are useful only when compressing files bigger than 8 MiB, 16 MiB, and 32 MiB, respectively.

On the same hardware, the decompression speed is approximately a constant number of bytes of compressed data per second. In other words, the better the compression, the faster the decompression will usually be. This also means that the amount of uncompressed output produced per second can vary a lot.

The following table summarises the features of the presets:

Preset	DictSize	CompCPU	CompMem	DecMem
-0	256 KiB	0	3 MiB	1 MiB
-1	1 MiB	1	9 MiB	2 MiB
-2	2 MiB	2	17 MiB	3 MiB
-3	4 MiB	3	32 MiB	5 MiB
-4	4 MiB	4	48 MiB	5 MiB
-5	8 MiB	5	94 MiB	9 MiB
-6	8 MiB	6	94 MiB	9 MiB
-7	16 MiB	6	186 MiB	17 MiB
-8	32 MiB	6	370 MiB	33 MiB
-9	64 MiB	6	674 MiB	65 MiB

Column descriptions:

- DictSize is the LZMA2 dictionary size. It is waste of memory to use a dictionary bigger than the size of the uncompressed file. This is why it is good to avoid using the presets **-7 ... -9** when there's no real need for them. At **-6** and lower, the amount of memory wasted is usually low enough to not matter.
- CompCPU is a simplified representation of the LZMA2 settings that affect compression speed. The dictionary size affects speed too, so while CompCPU is the same for levels **-6 ... -9**, higher levels still tend to be a little slower. To get even slower and thus possibly better compression, see **--extreme**.
- CompMem contains the compressor memory requirements in the single-threaded mode. It may vary slightly between **xz** versions.
- DecMem contains the decompressor memory requirements. That is, the compression settings determine the memory requirements of the decompressor. The exact decompressor memory usage is slightly more than the LZMA2 dictionary size, but the values in the table have been rounded up to the next full MiB.

Memory requirements of the multi-threaded mode are significantly higher than that of the single-threaded mode. With the default value of **--block-size**, each thread needs $3 \times 3 \times \text{DictSize}$ plus CompMem or DecMem. For example, four threads with preset **-6**

needs 660–670 MiB of memory.

-e, --extreme

Use a slower variant of the selected compression preset level (**-0** ... **-9**) to hopefully get a little bit better compression ratio, but with bad luck this can also make it worse. Decompressor memory usage is not affected, but compressor memory usage increases a little at preset levels **-0** ... **-3**.

Since there are two presets with dictionary sizes 4 MiB and 8 MiB, the presets **-3e** and **-5e** use slightly faster settings (lower CompCPU) than **-4e** and **-6e**, respectively. That way no two presets are identical.

Preset	DictSize	CompCPU	CompMem	DecMem
-0e	256 KiB	8	4 MiB	1 MiB
-1e	1 MiB	8	13 MiB	2 MiB
-2e	2 MiB	8	25 MiB	3 MiB
-3e	4 MiB	7	48 MiB	5 MiB
-4e	4 MiB	8	48 MiB	5 MiB
-5e	8 MiB	7	94 MiB	9 MiB
-6e	8 MiB	8	94 MiB	9 MiB
-7e	16 MiB	8	186 MiB	17 MiB
-8e	32 MiB	8	370 MiB	33 MiB
-9e	64 MiB	8	674 MiB	65 MiB

For example, there are a total of four presets that use 8 MiB dictionary, whose order from the fastest to the slowest is **-5**, **-6**, **-5e**, and **-6e**.

--fast

--best

These are somewhat misleading aliases for **-0** and **-9**, respectively. These are provided only for backwards compatibility with LZMA Utils. Avoid using these options.

--block-size=size

When compressing to the **.xz** format, split the input data into blocks of *size* bytes. The blocks are compressed independently from each other, which helps with multi-threading and makes limited random-access decompression possible. This option is typically used to override the default block size in multi-threaded mode, but this option can be used in single-threaded mode too.

In multi-threaded mode about three times *size* bytes will be allocated in each thread for buffering input and output. The default *size* is three times the LZMA2 dictionary size or 1 MiB, whichever is more. Typically a good value is 2–4 times the size of the LZMA2 dictionary or at least 1 MiB. Using *size* less than the LZMA2 dictionary size is waste of RAM because then the LZMA2 dictionary buffer will never get fully used. In multi-threaded mode, the sizes of the blocks are stored in the block headers. This size information is required for multi-threaded decompression.

In single-threaded mode no block splitting is done by default. Setting this option doesn't affect memory usage. No size information is stored in block headers, thus files created in single-threaded mode won't be identical to files created in multi-threaded mode. The lack of size information also means that **xz** won't be able decompress the files in multi-threaded mode.

--block-list=items

When compressing to the **.xz** format, start a new block with an optional custom filter chain after the given intervals of uncompressed data.

The *items* are a comma-separated list. Each item consists of an optional filter chain number between 0 and 9 followed by a colon (:) and a required size of uncompressed data. Omitting an item (two or more consecutive commas) is a shorthand to use the size and filters of the previous item.

If the input file is bigger than the sum of the sizes in *items*, the last item is repeated until the end of the file. A special value of **0** may be used as the last size to indicate that the rest of the file should be encoded as a single block.

An alternative filter chain for each block can be specified in combination with the **--filters1=filters ... --filters9=filters** options. These options define filter chains with an identifier between 1–9. Filter chain 0 can be used to refer to the default filter chain, which is the same as not specifying a filter chain. The filter chain identifier can be used before the uncompressed size, followed by a colon (:). For example, if one specifies **--block-list=1:2MiB,3:2MiB,2:4MiB,,2MiB,0:4MiB** then blocks will be created using:

- The filter chain specified by **--filters1** and 2 MiB input
- The filter chain specified by **--filters3** and 2 MiB input
- The filter chain specified by **--filters2** and 4 MiB input
- The filter chain specified by **--filters2** and 4 MiB input
- The default filter chain and 2 MiB input
- The default filter chain and 4 MiB input for every block until end of input.

If one specifies a size that exceeds the encoder's block size (either the default value in threaded mode or the value specified with **--block-size=size**), the encoder will create additional blocks while keeping the boundaries specified in *items*. For example, if one specifies **--block-size=10MiB --block-list=5MiB,10MiB,8MiB,12MiB,24MiB** and the input file is 80 MiB, one will get 11 blocks: 5, 10, 8, 10, 2, 10, 10, 4, 10, 10, and 1 MiB.

In multi-threaded mode the sizes of the blocks are stored in the block headers. This isn't done in single-threaded mode, so the encoded output won't be identical to that of the multi-threaded mode.

--flush-timeout=timeout

When compressing, if more than *timeout* milliseconds (a positive integer) has passed since the previous flush and reading more input would block, all the pending input data is flushed from the encoder and made available in the output stream. This can be useful if **xz** is used to compress data that is streamed over a network. Small *timeout* values make the data available at the receiving end with a small delay, but large *timeout* values give better compression ratio.

This feature is disabled by default. If this option is specified more than once, the last one takes effect. The special *timeout* value of **0** can be used to explicitly disable this feature.

This feature is not available on non-POSIX systems.

This feature is still experimental. Currently **xz** is unsuitable for decompressing the stream in real time due to how **xz** does buffering.

--no-sync

Do not synchronize the target file and its directory to the storage device before removing the source file. This can improve performance if compressing or decompressing many small files. However, if the system crashes soon after the deletion, it is possible that the target file was not written to the storage device but the delete operation was. In that case neither the original source file nor the target file is available.

This option has an effect only when **xz** is going to remove the source file. In other cases synchronization is never done.

The synchronization and **--no-sync** were added in **xz** 5.7.1alpha.

--memlimit-compress=limit

Set a memory usage limit for compression. If this option is specified multiple times, the last one takes effect.

If the compression settings exceed the *limit*, **xz** will attempt to adjust the settings downwards so that the limit is no longer exceeded and display a notice that automatic adjustment was done. The adjustments are done in this order: reducing the number of threads, switching to single-threaded mode if even one thread in multi-threaded mode exceeds the *limit*, and finally reducing the LZMA2 dictionary size.

When compressing with **--format=raw** or if **--no-adjust** has been specified, only the number of threads may be reduced since it can be done without affecting the compressed output.

If the *limit* cannot be met even with the adjustments described above, an error is displayed and **xz** will exit with exit status 1.

The *limit* can be specified in multiple ways:

- The *limit* can be an absolute value in bytes. Using an integer suffix like **MiB** can be useful. Example: **--memlimit-compress=80MiB**
- The *limit* can be specified as a percentage of total physical memory (RAM). This can be useful especially when setting the **XZ_DEFAULTS** environment variable in a shell initialization script that is shared between different computers. That way the limit is automatically bigger on systems with more memory. Example: **--memlimit-compress=70%**
- The *limit* can be reset back to its default value by setting it to **0**. This is currently equivalent to setting the *limit* to **max** (no memory usage limit).

For 32-bit **xz** there is a special case: if the *limit* would be over **4020 MiB**, the *limit* is set to **4020 MiB**. On MIPS32 **2000 MiB** is used instead. (The values **0** and **max** aren't affected by this. A similar feature doesn't exist for decompression.) This can

be helpful when a 32-bit executable has access to 4 GiB address space (2 GiB on MIPS32) while hopefully doing no harm in other situations.

See also the section **Memory usage**.

--memlimit-decompress=limit

Set a memory usage limit for decompression. This also affects the **--list** mode. If the operation is not possible without exceeding the *limit*, **xz** will display an error and decompressing the file will fail. See **--memlimit-compress=limit** for possible ways to specify the *limit*.

--memlimit-mt-decompress=limit

Set a memory usage limit for multi-threaded decompression. This can only affect the number of threads; this will never make **xz** refuse to decompress a file. If *limit* is too low to allow any multi-threading, the *limit* is ignored and **xz** will continue in single-threaded mode. Note that if also **--memlimit-decompress** is used, it will always apply to both single-threaded and multi-threaded modes, and so the effective *limit* for multi-threading will never be higher than the limit set with **--memlimit-decompress**.

In contrast to the other memory usage limit options, **--memlimit-mt-decompress=limit** has a system-specific default *limit*. **xz --info-memory** can be used to see the current value.

This option and its default value exist because without any limit the threaded decompressor could end up allocating an insane amount of memory with some input files. If the default *limit* is too low on your system, feel free to increase the *limit* but never set it to a value larger than the amount of usable RAM as with appropriate input files **xz** will attempt to use that amount of memory even with a low number of threads. Running out of memory or swapping will not improve decompression performance.

See **--memlimit-compress=limit** for possible ways to specify the *limit*. Setting *limit* to **0** resets the *limit* to the default system-specific value.

-M limit, --memlimit=limit, --memory=limit

This is equivalent to specifying **--memlimit-compress=limit --memlimit-decompress=limit --memlimit-mt-decompress=limit**.

--no-adjust

Display an error and exit if the memory usage limit cannot be met without adjusting settings that affect the compressed output. That is, this prevents **xz** from switching the encoder from multi-threaded mode to single-threaded mode and from reducing the LZMA2 dictionary size. Even when this option is used the number of threads may be reduced to meet the memory usage limit as that won't affect the compressed output.

Automatic adjusting is always disabled when creating raw streams (**--format=raw**).

-T threads, --threads=threads

Specify the number of worker threads to use. Setting *threads* to a special value **0** makes **xz** use up to as many threads as the processor(s) on the system support. The actual number of threads can be fewer than *threads* if the input file is not big enough for threading with the given settings or if using more threads would exceed the memory usage limit.

The single-threaded and multi-threaded compressors produce different output. Single-threaded compressor will give the smallest file size but only the output from the multi-threaded compressor can be decompressed using multiple threads. Setting *threads* to **1** will use the single-threaded mode. Setting *threads* to any other value, including **0**, will use the multi-threaded compressor even if the system supports only one hardware thread. (**xz** 5.2.x used single-threaded mode in this situation.)

To use multi-threaded mode with only one thread, set *threads* to **+1**. The **+** prefix has no effect with values other than **1**. A memory usage limit can still make **xz** switch to single-threaded mode unless **--no-adjust** is used. Support for the **+** prefix was added in **xz** 5.4.0.

If an automatic number of threads has been requested and no memory usage limit has been specified, then a system-specific default soft limit will be used to possibly limit the number of threads. It is a soft limit in sense that it is ignored if the number of threads becomes one, thus a soft limit will never stop **xz** from compressing or decompressing. This default soft limit will not make **xz** switch from multi-threaded mode to single-threaded mode. The active limits can be seen with **xz --info-memory**.

Currently the only threading method is to split the input into blocks and compress them independently from each other. The default block size depends on the compression level and can be overridden with the **--block-size=size** option.

Threaded decompression only works on files that contain multiple blocks with size information in block headers. All large enough files compressed in multi-threaded mode meet this condition, but files compressed in single-threaded mode don't even if **--block-size=size** has been used.

The default value for *threads* is **0**. In **xz** 5.4.x and older the default is **1**.

Custom compressor filter chains

A custom filter chain allows specifying the compression settings in detail instead of relying on the settings associated to the presets. When a custom filter chain is specified, preset options (**-0 ... -9** and **--extreme**) earlier on the command line are forgotten. If a preset option is specified after one or more custom filter chain options, the new preset takes effect and the custom filter chain options specified earlier are forgotten.

A filter chain is comparable to piping on the command line. When compressing, the uncompressed input goes to the first filter, whose output goes to the next filter (if any). The output of the last filter gets written to the compressed file. The maximum number of filters in the chain is four, but typically a filter chain has only one or two filters.

Many filters have limitations on where they can be in the filter chain: some filters can work only as the last filter in the chain, some only as a non-last filter, and some work in any position in the chain. Depending on the filter, this limitation is either inherent to the filter design or exists to prevent security issues.

A custom filter chain can be specified in two different ways. The options **--filters=filters** and **--filters1=filters ... --filters9=filters** allow specifying an entire filter chain in one option using the liblzma filter string syntax. Alternatively, a filter chain can be specified by using one or more individual filter options in the order they are wanted in the filter chain. That is, the order of the individual filter options is significant! When decoding raw streams (**--format=raw**), the filter chain must be specified in the same order as it was specified when compressing. Any individual filter or preset options specified before the full chain option

(**--filters=***filters*) will be forgotten. Individual filters specified after the full chain option will reset the filter chain.

Both the full and individual filter options take filter-specific *options* as a comma-separated list. Extra commas in *options* are ignored. Every option has a default value, so specify those you want to change.

To see the whole filter chain and *options*, use **xz -vv** (that is, use **--verbose** twice). This works also for viewing the filter chain options used by presets.

--filters=*filters*

Specify the full filter chain or a preset in a single option. Each filter can be separated by spaces or two dashes (**--**). *filters* may need to be quoted on the shell command line so it is parsed as a single option. To denote *options*, use **:** or **=**. A preset can be prefixed with a **-** and followed with zero or more flags. The only supported flag is **e** to apply the same options as **--extreme**.

--filters1=*filters* ... **--filters9=***filters*

Specify up to nine additional filter chains that can be used with **--block-list**.

For example, when compressing an archive with executable files followed by text files, the executable part could use a filter chain with a BCJ filter and the text part only the LZMA2 filter.

--filters-help

Display a help message describing how to specify presets and custom filter chains in the **--filters** and **--filters1=***filters* ... **--filters9=***filters* options, and exit successfully.

--lzma1[=*options***]**

--lzma2[=*options***]**

Add LZMA1 or LZMA2 filter to the filter chain. These filters can be used only as the last filter in the chain.

LZMA1 is a legacy filter, which is supported almost solely due to the legacy **.lzma** file format, which supports only LZMA1. LZMA2 is an updated version of LZMA1 to fix some practical issues of LZMA1. The **.xz** format uses LZMA2 and doesn't support LZMA1 at all. Compression speed and ratios of LZMA1 and LZMA2 are practically the same.

LZMA1 and LZMA2 share the same set of *options*:

preset=*preset*

Reset all LZMA1 or LZMA2 *options* to *preset*. *Preset* consist of an integer, which may be followed by single-letter preset modifiers. The integer can be from **0** to **9**, matching the command line options **-0** ... **-9**. The only supported modifier is currently **e**, which matches **--extreme**. If no **preset** is specified, the default values of LZMA1 or LZMA2 *options* are taken from the preset **6**.

dict=*size*

Dictionary (history buffer) *size* indicates how many bytes of the recently processed uncompressed data is kept in memory. The algorithm tries to find repeating byte sequences (matches) in the uncompressed data, and replace

them with references to the data currently in the dictionary. The bigger the dictionary, the higher is the chance to find a match. Thus, increasing dictionary *size* usually improves compression ratio, but a dictionary bigger than the uncompressed file is waste of memory.

Typical dictionary *size* is from 64 KiB to 64 MiB. The minimum is 4 KiB. The maximum for compression is currently 1.5 GiB (1536 MiB). The decompressor already supports dictionaries up to one byte less than 4 GiB, which is the maximum for the LZMA1 and LZMA2 stream formats.

Dictionary *size* and match finder (*mf*) together determine the memory usage of the LZMA1 or LZMA2 encoder. The same (or bigger) dictionary *size* is required for decompressing that was used when compressing, thus the memory usage of the decoder is determined by the dictionary size used when compressing. The **.xz** headers store the dictionary *size* either as 2^n or $2^n + 2^{(n-1)}$, so these *sizes* are somewhat preferred for compression. Other *sizes* will get rounded up when stored in the **.xz** headers.

lc=lc Specify the number of literal context bits. The minimum is 0 and the maximum is 4; the default is 3. In addition, the sum of *lc* and *lp* must not exceed 4.

All bytes that cannot be encoded as matches are encoded as literals. That is, literals are simply 8-bit bytes that are encoded one at a time.

The literal coding makes an assumption that the highest *lc* bits of the previous uncompressed byte correlate with the next byte. For example, in typical English text, an upper-case letter is often followed by a lower-case letter, and a lower-case letter is usually followed by another lower-case letter. In the US-ASCII character set, the highest three bits are 010 for upper-case letters and 011 for lower-case letters. When *lc* is at least 3, the literal coding can take advantage of this property in the uncompressed data.

The default value (3) is usually good. If you want maximum compression, test **lc=4**. Sometimes it helps a little, and sometimes it makes compression worse. If it makes it worse, test **lc=2** too.

lp=lp Specify the number of literal position bits. The minimum is 0 and the maximum is 4; the default is 0.

Lp affects what kind of alignment in the uncompressed data is assumed when encoding literals. See *pb* below for more information about alignment.

pb=pb

Specify the number of position bits. The minimum is 0 and the maximum is 4; the default is 2.

Pb affects what kind of alignment in the uncompressed data is assumed in general. The default means four-byte alignment ($2^{pb}=2^2=4$), which is often a good choice when there's no better guess.

When the alignment is known, setting *pb* accordingly may reduce the file size a little. For example, with text files having one-byte alignment (US-ASCII, ISO-8859-*, UTF-8), setting **pb=0** can improve compression slightly. For

UTF-16 text, **pb=1** is a good choice. If the alignment is an odd number like 3 bytes, **pb=0** might be the best choice.

Even though the assumed alignment can be adjusted with *pb* and *lp*, LZMA1 and LZMA2 still slightly favor 16-byte alignment. It might be worth taking into account when designing file formats that are likely to be often compressed with LZMA1 or LZMA2.

mf=*mf*

Match finder has a major effect on encoder speed, memory usage, and compression ratio. Usually Hash Chain match finders are faster than Binary Tree match finders. The default depends on the *preset*: 0 uses **hc3**, 1–3 use **hc4**, and the rest use **bt4**.

The following match finders are supported. The memory usage formulas below are rough approximations, which are closest to the reality when *dict* is a power of two.

- hc3** Hash Chain with 2- and 3-byte hashing
Minimum value for *nice*: 3
Memory usage:
dict * 7.5 (if *dict* ≤ 16 MiB);
dict * 5.5 + 64 MiB (if *dict* > 16 MiB)
- hc4** Hash Chain with 2-, 3-, and 4-byte hashing
Minimum value for *nice*: 4
Memory usage:
dict * 7.5 (if *dict* ≤ 32 MiB);
dict * 6.5 (if *dict* > 32 MiB)
- bt2** Binary Tree with 2-byte hashing
Minimum value for *nice*: 2
Memory usage: *dict* * 9.5
- bt3** Binary Tree with 2- and 3-byte hashing
Minimum value for *nice*: 3
Memory usage:
dict * 11.5 (if *dict* ≤ 16 MiB);
dict * 9.5 + 64 MiB (if *dict* > 16 MiB)
- bt4** Binary Tree with 2-, 3-, and 4-byte hashing
Minimum value for *nice*: 4
Memory usage:
dict * 11.5 (if *dict* ≤ 32 MiB);
dict * 10.5 (if *dict* > 32 MiB)

mode=*mode*

Compression *mode* specifies the method to analyze the data produced by the match finder. Supported *modes* are **fast** and **normal**. The default is **fast** for *presets* 0–3 and **normal** for *presets* 4–9.

Usually **fast** is used with Hash Chain match finders and **normal** with Binary Tree match finders. This is also what the *presets* do.

nice=*nice*

Specify what is considered to be a nice length for a match. Once a match of at least *nice* bytes is found, the algorithm stops looking for possibly better matches.

Nice can be 2–273 bytes. Higher values tend to give better compression ratio at the expense of speed. The default depends on the *preset*.

depth=*depth*

Specify the maximum search depth in the match finder. The default is the special value of 0, which makes the compressor determine a reasonable *depth* from *mf* and *nice*.

Reasonable *depth* for Hash Chains is 4–100 and 16–1000 for Binary Trees. Using very high values for *depth* can make the encoder extremely slow with some files. Avoid setting the *depth* over 1000 unless you are prepared to interrupt the compression in case it is taking far too long.

When decoding raw streams (**--format=raw**), LZMA2 needs only the dictionary *size*. LZMA1 needs also *lc*, *lp*, and *pb*.

--x86[=*options*]

--arm[=*options*]

--armthumb[=*options*]

--arm64[=*options*]

--powerpc[=*options*]

--ia64[=*options*]

--sparc[=*options*]

--riscv[=*options*]

Add a branch/call/jump (BCJ) filter to the filter chain. These filters can be used only as a non-last filter in the filter chain.

A BCJ filter converts relative addresses in the machine code to their absolute counterparts. This doesn't change the size of the data but it increases redundancy, which can help LZMA2 to produce 0–15 % smaller **.xz** file. The BCJ filters are always reversible, so using a BCJ filter for wrong type of data doesn't cause any data loss, although it may make the compression ratio slightly worse. The BCJ filters are very fast and use an insignificant amount of memory.

These BCJ filters have known problems related to the compression ratio:

- Some types of files containing executable code (for example, object files, static libraries, and Linux kernel modules) have the addresses in the instructions filled with filler values. These BCJ filters will still do the address conversion, which will make the compression worse with these files.
- If a BCJ filter is applied on an archive, it is possible that it makes the compression ratio worse than not using a BCJ filter. For example, if there are similar or even identical executables then filtering will likely make the files less similar and thus compression is worse. The contents of non-executable files in the same archive can matter too. In practice one has to try with and without a BCJ filter to see which is better in each situation.

Different instruction sets have different alignment: the executable file must be aligned to a multiple of this value in the input data to make the filter work.

Filter	Alignment	Notes
x86	1	32-bit or 64-bit x86
ARM	4	
ARM-Thumb	2	
ARM64	4	4096-byte alignment is best
PowerPC	4	Big endian only
IA-64	16	Itanium
SPARC	4	
RISC-V	2	

Since the BCJ-filtered data is usually compressed with LZMA2, the compression ratio may be improved slightly if the LZMA2 options are set to match the alignment of the selected BCJ filter. Examples:

- IA-64 filter has 16-byte alignment so **pb=4,lp=4,lc=0** is good with LZMA2 ($2^4=16$).
- RISC-V code has 2-byte or 4-byte alignment depending on whether the file contains 16-bit compressed instructions (the C extension). When 16-bit instructions are used, **pb=2,lp=1,lc=3** or **pb=1,lp=1,lc=3** is good. When 16-bit instructions aren't present, **pb=2,lp=2,lc=2** is the best. **readelf -h** can be used to check if "RVC" appears on the "Flags" line.
- ARM64 is always 4-byte aligned so **pb=2,lp=2,lc=2** is the best.
- The x86 filter is an exception. It's usually good to stick to LZMA2's defaults (**pb=2,lp=0,lc=3**) when compressing x86 executables.

All BCJ filters support the same *options*:

start=offset

Specify the start *offset* that is used when converting between relative and absolute addresses. The *offset* must be a multiple of the alignment of the filter (see the table above). The default is zero. In practice, the default is good; specifying a custom *offset* is almost never useful.

--delta[=options]

Add the Delta filter to the filter chain. The Delta filter can be only used as a non-last filter in the filter chain.

Currently only simple byte-wise delta calculation is supported. It can be useful when compressing, for example, uncompressed bitmap images or uncompressed PCM audio. However, special purpose algorithms may give significantly better results than Delta + LZMA2. This is true especially with audio, which compresses faster and better, for example, with **flac(1)**.

Supported *options*:

dist=distance

Specify the *distance* of the delta calculation in bytes. *distance* must be 1–256. The default is 1.

For example, with **dist=2** and eight-byte input A1 B1 A2 B3 A3 B5 A4 B7, the output will be A1 B1 01 02 01 02 01 02.

Other options

-q, --quiet

Suppress warnings and notices. Specify this twice to suppress errors too. This option has no effect on the exit status. That is, even if a warning was suppressed, the exit status to indicate a warning is still used.

-v, --verbose

Be verbose. If standard error is connected to a terminal, **xz** will display a progress indicator. Specifying **--verbose** twice will give even more verbose output.

The progress indicator shows the following information:

- Completion percentage is shown if the size of the input file is known. That is, the percentage cannot be shown in pipes.
- Amount of compressed data produced (compressing) or consumed (decompressing).
- Amount of uncompressed data consumed (compressing) or produced (decompressing).
- Compression ratio, which is calculated by dividing the amount of compressed data processed so far by the amount of uncompressed data processed so far.
- Compression or decompression speed. This is measured as the amount of uncompressed data consumed (compression) or produced (decompression) per second. It is shown after a few seconds have passed since **xz** started processing the file.
- Elapsed time in the format M:SS or H:MM:SS.
- Estimated remaining time is shown only when the size of the input file is known and a couple of seconds have already passed since **xz** started processing the file. The time is shown in a less precise format which never has any colons, for example, 2 min 30 s.

When standard error is not a terminal, **--verbose** will make **xz** print the filename, compressed size, uncompressed size, compression ratio, and possibly also the speed and elapsed time on a single line to standard error after compressing or decompressing the file. The speed and elapsed time are included only when the operation took at least a few seconds. If the operation didn't finish, for example, due to user interruption, also the completion percentage is printed if the size of the input file is known.

-Q, --no-warn

Don't set the exit status to 2 even if a condition worth a warning was detected. This option doesn't affect the verbosity level, thus both **--quiet** and **--no-warn** have to be used to not display warnings and to not alter the exit status.

--robot

Print messages in a machine-parsable format. This is intended to ease writing frontends that want to use **xz** instead of liblzma, which may be the case with various scripts. The output with this option enabled is meant to be stable across **xz** releases.

See the section **ROBOT MODE** for details.

--info-memory

Display, in human-readable format, how much physical memory (RAM) and how many processor threads **xz** thinks the system has and the memory usage limits for compression and decompression, and exit successfully.

-h, --help

Display a help message describing the most commonly used options, and exit successfully.

-H, --long-help

Display a help message describing all features of **xz**, and exit successfully

-V, --version

Display the version number of **xz** and liblzma in human readable format. To get machine-parsable output, specify **--robot** before **--version**.

ROBOT MODE

The robot mode is activated with the **--robot** option. It makes the output of **xz** easier to parse by other programs. Currently **--robot** is supported only together with **--list**, **--filters-help**, **--info-memory**, and **--version**. It will be supported for compression and decompression in the future.

List mode

xz --robot --list uses tab-separated output. The first column of every line has a string that indicates the type of the information found on that line:

name This is always the first line when starting to list a file. The second column on the line is the filename.

file This line contains overall information about the **.xz** file. This line is always printed after the **name** line.

stream

This line type is used only when **--verbose** was specified. There are as many **stream** lines as there are streams in the **.xz** file.

block This line type is used only when **--verbose** was specified. There are as many **block** lines as there are blocks in the **.xz** file. The **block** lines are shown after all the **stream** lines; different line types are not interleaved.

summary

This line type is used only when **--verbose** was specified twice. This line is printed after all **block** lines. Like the **file** line, the **summary** line contains overall information about the **.xz** file.

totals This line is always the very last line of the list output. It shows the total counts and sizes.

The columns of the **file** lines:

2. Number of streams in the file

3. Total number of blocks in the stream(s)
4. Compressed size of the file
5. Uncompressed size of the file
6. Compression ratio, for example, **0.123**. If ratio is over 9.999, three dashes (---) are displayed instead of the ratio.
7. Comma-separated list of integrity check names. The following strings are used for the known check types: **None**, **CRC32**, **CRC64**, and **SHA-256**. For unknown check types, **Unknown-N** is used, where *N* is the Check ID as a decimal number (one or two digits).
8. Total size of stream padding in the file

The columns of the **stream** lines:

2. Stream number (the first stream is 1)
3. Number of blocks in the stream
4. Compressed start offset
5. Uncompressed start offset
6. Compressed size (does not include stream padding)
7. Uncompressed size
8. Compression ratio
9. Name of the integrity check
10. Size of stream padding

The columns of the **block** lines:

2. Number of the stream containing this block
3. Block number relative to the beginning of the stream (the first block is 1)
4. Block number relative to the beginning of the file
5. Compressed start offset relative to the beginning of the file
6. Uncompressed start offset relative to the beginning of the file
7. Total compressed size of the block (includes headers)
8. Uncompressed size
9. Compression ratio
10. Name of the integrity check

If **--verbose** was specified twice, additional columns are included on the **block** lines. These are not displayed with a single **--verbose**, because getting this information requires many seeks and can thus be slow:

11. Value of the integrity check in hexadecimal
12. Block header size
13. Block flags: **c** indicates that compressed size is present, and **u** indicates that uncompressed size is present. If the flag is not set, a dash (–) is shown instead to keep the string length fixed. New flags may be added to the end of the string in the future.
14. Size of the actual compressed data in the block (this excludes the block header, block padding, and check fields)
15. Amount of memory (in bytes) required to decompress this block with this **xz** version
16. Filter chain. Note that most of the options used at compression time cannot be known, because only the options that are needed for decompression are stored in the **.xz** headers.

The columns of the **summary** lines:

2. Amount of memory (in bytes) required to decompress this file with this **xz** version
3. **yes** or **no** indicating if all block headers have both compressed size and uncompressed size stored in them

Since **xz 5.1.2alpha**:

4. Minimum **xz** version required to decompress the file

The columns of the **totals** line:

2. Number of streams
3. Number of blocks
4. Compressed size
5. Uncompressed size
6. Average compression ratio
7. Comma-separated list of integrity check names that were present in the files
8. Stream padding size
9. Number of files. This is here to keep the order of the earlier columns the same as on **file** lines.

If **--verbose** was specified twice, additional columns are included on the **totals** line:

10. Maximum amount of memory (in bytes) required to decompress the files with this **xz** version
11. **yes** or **no** indicating if all block headers have both compressed size and uncompressed size stored in them

Since **xz 5.1.2alpha**:

12. Minimum **xz** version required to decompress the file

Future versions may add new line types and new columns can be added to the existing line types, but the existing columns won't be changed.

Filters help

xz --robot --filters-help prints the supported filters in the following format:

filter:option=<value>,option=<value>...

filter Name of the filter

option Name of a filter specific option

value Numeric *value* ranges appear as *<min-max>*. String *value* choices are shown within *< >* and separated by a *|* character.

Each filter is printed on its own line.

Memory limit information

xz --robot --info-memory prints a single line with multiple tab-separated columns:

1. Total amount of physical memory (RAM) in bytes.
2. Memory usage limit for compression in bytes (**--memlimit-compress**). A special value of **0** indicates the default setting which for single-threaded mode is the same as no limit.
3. Memory usage limit for decompression in bytes (**--memlimit-decompress**). A special value of **0** indicates the default setting which for single-threaded mode is the same as no limit.

4. Since **xz** 5.3.4alpha: Memory usage for multi-threaded decompression in bytes (**--mem-limit-mt-decompress**). This is never zero because a system-specific default value shown in the column 5 is used if no limit has been specified explicitly. This is also never greater than the value in the column 3 even if a larger value has been specified with **--memlimit-mt-decompress**.
5. Since **xz** 5.3.4alpha: A system-specific default memory usage limit that is used to limit the number of threads when compressing with an automatic number of threads (**--threads=0**) and no memory usage limit has been specified (**--memlimit-compress**). This is also used as the default value for **--memlimit-mt-decompress**.
6. Since **xz** 5.3.4alpha: Number of available processor threads.

In the future, the output of **xz --robot --info-memory** may have more columns, but never more than a single line.

Version

xz --robot --version prints the version number of **xz** and liblzma in the following format:

```
XZ_VERSION=YYYYZZZS
LIBLZMA_VERSION=YYYYZZZS
```

X Major version.

YYY Minor version. Even numbers are stable. Odd numbers are alpha or beta versions.

ZZZ Patch level for stable releases or just a counter for development releases.

S Stability. 0 is alpha, 1 is beta, and 2 is stable. **S** should be always 2 when **YYY** is even.

YYYYZZZS are the same on both lines if **xz** and liblzma are from the same XZ Utils release.

Examples: 4.999.9beta is **49990091** and 5.0.0 is **50000002**.

EXIT STATUS

0 All is good.

1 An error occurred.

2 Something worth a warning occurred, but no actual errors occurred.

Notices (not warnings or errors) printed on standard error don't affect the exit status.

ENVIRONMENT

xz parses space-separated lists of options from the environment variables **XZ_DEFAULTS** and **XZ_OPT**, in this order, before parsing the options from the command line. Note that only options are parsed from the environment variables; all non-options are silently ignored. Parsing is done with **getopt_long(3)** which is used also for the command line arguments.

Warning: By setting these environment variables, one is effectively modifying programs and scripts that run **xz**. Most of the time it is safe to set memory usage limits, number of threads, and compression options via the environment variables. However, some options can break scripts. An obvious example is **--help** which makes **xz** show the help text instead of compressing or decompressing a file. More subtle examples are **--quiet** and **--verbose**. In many

cases it works well to enable the progress indicator using **--verbose**, but in some situations the extra messages create problems. The verbosity level also affects the behavior of **--list**.

XZ_DEFAULTS

User-specific or system-wide default options. Typically this is set in a shell initialization script to enable **xz**'s memory usage limiter by default or set the default number of threads. Excluding shell initialization scripts and similar special cases, scripts should never set or unset **XZ_DEFAULTS**.

XZ_OPT

This is for passing options to **xz** when it is not possible to set the options directly on the **xz** command line. This is the case when **xz** is run by a script or tool, for example, GNU **tar**(1):

```
XZ_OPT=-2v tar caf foo.tar.xz foo
```

Scripts may use **XZ_OPT**, for example, to set script-specific default compression options. It is still recommended to allow users to override **XZ_OPT** if that is reasonable. For example, in **sh**(1) scripts one may use something like this:

```
XZ_OPT=${XZ_OPT-"-7e"}
export XZ_OPT
```

LZMA UTILS COMPATIBILITY

The command line syntax of **xz** is practically a superset of **lzma**, **unlzma**, and **lzc** as found from LZMA Utils 4.32.x. In most cases, it is possible to replace LZMA Utils with XZ Utils without breaking existing scripts. There are some incompatibilities though, which may sometimes cause problems.

Compression preset levels

The numbering of the compression level presets is not identical in **xz** and LZMA Utils. The most important difference is how dictionary sizes are mapped to different presets. Dictionary size is roughly equal to the decompressor memory usage.

Level	xz	LZMA Utils
-0	256 KiB	N/A
-1	1 MiB	64 KiB
-2	2 MiB	1 MiB
-3	4 MiB	512 KiB
-4	4 MiB	1 MiB
-5	8 MiB	2 MiB
-6	8 MiB	4 MiB
-7	16 MiB	8 MiB
-8	32 MiB	16 MiB
-9	64 MiB	32 MiB

The dictionary size differences affect the compressor memory usage too, but there are some other differences between LZMA Utils and XZ Utils, which make the difference even bigger:

Level	xz	LZMA Utils 4.32.x
-0	3 MiB	N/A
-1	9 MiB	2 MiB
-2	17 MiB	12 MiB

-3	32 MiB	12 MiB
-4	48 MiB	16 MiB
-5	94 MiB	26 MiB
-6	94 MiB	45 MiB
-7	186 MiB	83 MiB
-8	370 MiB	159 MiB
-9	674 MiB	311 MiB

The default preset level in LZMA Utils is **-7** while in XZ Utils it is **-6**, so both use an 8 MiB dictionary by default.

Streamed vs. non-streamed .lzma files

The uncompressed size of the file can be stored in the **.lzma** header. LZMA Utils does that when compressing regular files. The alternative is to mark that uncompressed size is unknown and use end-of-payload marker to indicate where the decompressor should stop. LZMA Utils uses this method when uncompressed size isn't known, which is the case, for example, in pipes.

xz supports decompressing **.lzma** files with or without end-of-payload marker, but all **.lzma** files created by **xz** will use end-of-payload marker and have uncompressed size marked as unknown in the **.lzma** header. This may be a problem in some uncommon situations. For example, a **.lzma** decompressor in an embedded device might work only with files that have known uncompressed size. If you hit this problem, you need to use LZMA Utils or LZMA SDK to create **.lzma** files with known uncompressed size.

Unsupported .lzma files

The **.lzma** format allows *lc* values up to 8, and *lp* values up to 4. LZMA Utils can decompress files with any *lc* and *lp*, but always creates files with **lc=3** and **lp=0**. Creating files with other *lc* and *lp* is possible with **xz** and with LZMA SDK.

The implementation of the LZMA1 filter in liblzma requires that the sum of *lc* and *lp* must not exceed 4. Thus, **.lzma** files, which exceed this limitation, cannot be decompressed with **xz**.

LZMA Utils creates only **.lzma** files which have a dictionary size of 2^n (a power of 2) but accepts files with any dictionary size. liblzma accepts only **.lzma** files which have a dictionary size of 2^n or $2^n + 2^{(n-1)}$. This is to decrease false positives when detecting **.lzma** files.

These limitations shouldn't be a problem in practice, since practically all **.lzma** files have been compressed with settings that liblzma will accept.

Trailing garbage

When decompressing, LZMA Utils silently ignore everything after the first **.lzma** stream. In most situations, this is a bug. This also means that LZMA Utils don't support decompressing concatenated **.lzma** files.

If there is data left after the first **.lzma** stream, **xz** considers the file to be corrupt unless **--single-stream** was used. This may break obscure scripts which have assumed that trailing garbage is ignored.

NOTES

Compressed output may vary

The exact compressed output produced from the same uncompressed input file may vary between XZ Utils versions even if compression options are identical. This is because the encoder can be improved (faster or better compression) without affecting the file format. The

output can vary even between different builds of the same XZ Utils version, if different build options are used.

The above means that once **--rsyncable** has been implemented, the resulting files won't necessarily be rsyncable unless both old and new files have been compressed with the same xz version. This problem can be fixed if a part of the encoder implementation is frozen to keep rsyncable output stable across xz versions.

Embedded .xz decompressors

Embedded .xz decompressor implementations like XZ Embedded don't necessarily support files created with integrity *check* types other than **none** and **crc32**. Since the default is **--check=crc64**, you must use **--check=none** or **--check=crc32** when creating files for embedded systems.

Outside embedded systems, all .xz format decompressors support all the *check* types, or at least are able to decompress the file without verifying the integrity check if the particular *check* is not supported.

XZ Embedded supports BCJ filters, but only with the default start offset.

EXAMPLES

Basics

Compress the file *foo* into *foo.xz* using the default compression level (**-6**), and remove *foo* if compression is successful:

```
xz foo
```

Decompress *bar.xz* into *bar* and don't remove *bar.xz* even if decompression is successful:

```
xz -dk bar.xz
```

Create *baz.tar.xz* with the preset **-4e** (**-4 --extreme**), which is slower than the default **-6**, but needs less memory for compression and decompression (48 MiB and 5 MiB, respectively):

```
tar cf - baz | xz -4e > baz.tar.xz
```

A mix of compressed and uncompressed files can be decompressed to standard output with a single command:

```
xz -dcf a.txt b.txt.xz c.txt d.txt.lzma > abcd.txt
```

Parallel compression of many files

On GNU and *BSD, **find(1)** and **xargs(1)** can be used to parallelize compression of many files:

```
find . -type f \! -name '*.xz' -print0 \
    | xargs -0r -P4 -n16 xz -T1
```

The **-P** option to **xargs(1)** sets the number of parallel **xz** processes. The best value for the **-n** option depends on how many files there are to be compressed. If there are only a couple of files, the value should probably be 1; with tens of thousands of files, 100 or even more may be appropriate to reduce the number of **xz** processes that **xargs(1)** will eventually create.

The option **-T1** for **xz** is there to force it to single-threaded mode, because **xargs(1)** is used to control the amount of parallelization.

Robot mode

Calculate how many bytes have been saved in total after compressing multiple files:

```
xz --robot --list *.xz | awk '/^totals/{print $5-$4}'
```

A script may want to know that it is using new enough **xz**. The following **sh**(1) script checks that the version number of the **xz** tool is at least 5.0.0. This method is compatible with old beta versions, which didn't support the **--robot** option:

```
if ! eval "$(xz --robot --version 2> /dev/null)" ||
    [ "$XZ_VERSION" -lt 50000002 ]; then
    echo "Your xz is too old."
fi
unset XZ_VERSION LIBLZMA_VERSION
```

Set a memory usage limit for decompression using **XZ_OPT**, but if a limit has already been set, don't increase it:

```
NEWLIM=$((123 << 20)) # 123 MiB
OLDLIM=$(xz --robot --info-memory | cut -f3)
if [ $OLDLIM -eq 0 -o $OLDLIM -gt $NEWLIM ]; then
    XZ_OPT="$XZ_OPT --memlimit-decompress=$NEWLIM"
    export XZ_OPT
fi
```

Custom compressor filter chains

The simplest use for custom filter chains is customizing a LZMA2 preset. This can be useful, because the presets cover only a subset of the potentially useful combinations of compression settings.

The CompCPU columns of the tables from the descriptions of the options **-0** ... **-9** and **--extreme** are useful when customizing LZMA2 presets. Here are the relevant parts collected from those two tables:

Preset	CompCPU
-0	0
-1	1
-2	2
-3	3
-4	4
-5	5
-6	6
-5e	7
-6e	8

If you know that a file requires somewhat big dictionary (for example, 32 MiB) to compress well, but you want to compress it quicker than **xz -8** would do, a preset with a low CompCPU value (for example, 1) can be modified to use a bigger dictionary:

```
xz --lzma2=preset=1,dict=32MiB foo.tar
```

With certain files, the above command may be faster than **xz -6** while compressing significantly better. However, it must be emphasized that only some files benefit from a big dictionary while keeping the CompCPU value low. The most obvious situation, where a big

dictionary can help a lot, is an archive containing very similar files of at least a few megabytes each. The dictionary size has to be significantly bigger than any individual file to allow LZMA2 to take full advantage of the similarities between consecutive files.

If very high compressor and decompressor memory usage is fine, and the file being compressed is at least several hundred megabytes, it may be useful to use an even bigger dictionary than the 64 MiB that **xz -9** would use:

```
xz -vv --lzma2=dict=192MiB big_foo.tar
```

Using **-vv** (**--verbose --verbose**) like in the above example can be useful to see the memory requirements of the compressor and decompressor. Remember that using a dictionary bigger than the size of the uncompressed file is waste of memory, so the above command isn't useful for small files.

Sometimes the compression time doesn't matter, but the decompressor memory usage has to be kept low, for example, to make it possible to decompress the file on an embedded system. The following command uses **-6e** (**-6 --extreme**) as a base and sets the dictionary to only 64 KiB. The resulting file can be decompressed with XZ Embedded (that's why there is **--check=crc32**) using about 100 KiB of memory.

```
xz --check=crc32 --lzma2=preset=6e,dict=64KiB foo
```

If you want to squeeze out as many bytes as possible, adjusting the number of literal context bits (*lc*) and number of position bits (*pb*) can sometimes help. Adjusting the number of literal position bits (*lp*) might help too, but usually *lc* and *pb* are more important. For example, a source code archive contains mostly US-ASCII text, so something like the following might give slightly (like 0.1 %) smaller file than **xz -6e** (try also without **lc=4**):

```
xz --lzma2=preset=6e,pb=0,lc=4 source_code.tar
```

Using another filter together with LZMA2 can improve compression with certain file types. For example, to compress a x86-32 or x86-64 shared library using the x86 BCJ filter:

```
xz --x86 --lzma2 libfoo.so
```

Note that the order of the filter options is significant. If **--x86** is specified after **--lzma2**, **xz** will give an error, because there cannot be any filter after LZMA2, and also because the x86 BCJ filter cannot be used as the last filter in the chain.

The Delta filter together with LZMA2 can give good results with bitmap images. It should usually beat PNG, which has a few more advanced filters than simple delta but uses Deflate for the actual compression.

The image has to be saved in uncompressed format, for example, as uncompressed TIFF. The distance parameter of the Delta filter is set to match the number of bytes per pixel in the image. For example, 24-bit RGB bitmap needs **dist=3**, and it is also good to pass **pb=0** to LZMA2 to accommodate the three-byte alignment:

```
xz --delta=dist=3 --lzma2=pb=0 foo.tiff
```

If multiple images have been put into a single archive (for example, **.tar**), the Delta filter will work on that too as long as all images have the same number of bytes per pixel.

SEE ALSO

xzdec(1), xzdiff(1), xzgrep(1), xzless(1), xzmore(1), gzip(1), bzip2(1), 7z(1)

XZ Utils: <<https://tukaani.org/xz/>>

XZ Embedded: <<https://tukaani.org/xz/embedded.html>>

LZMA SDK: <<https://7-zip.org/sdk.html>>